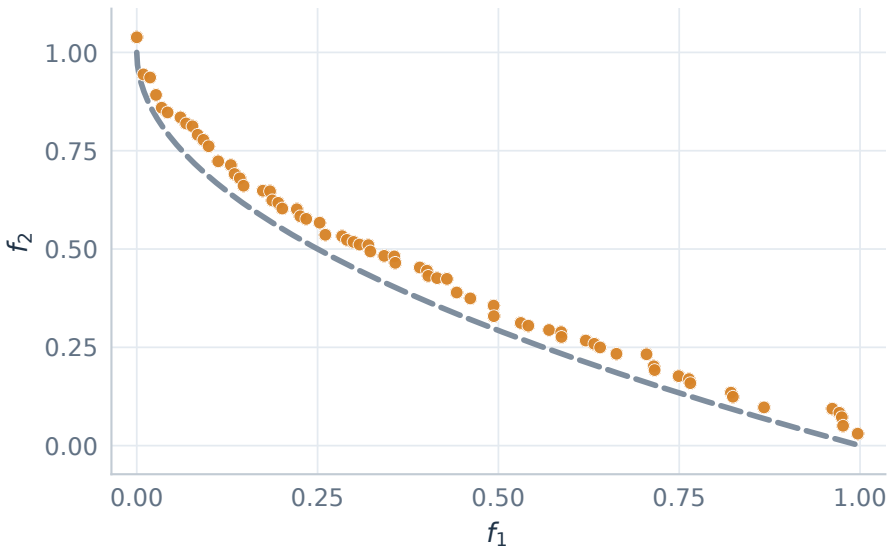


RVEA family / three simpler two-objective problems

ZDT1: convex | ZDT2: concave | ZDT3: disconnected. 30 decision variables in each problem.

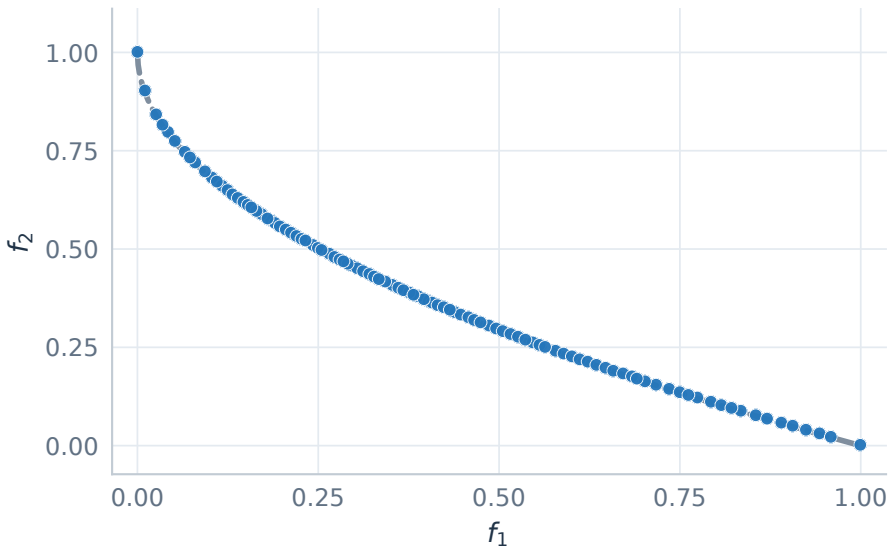
ZDT1 / RVEA

68 nondominated points | IGD+ 0.038176



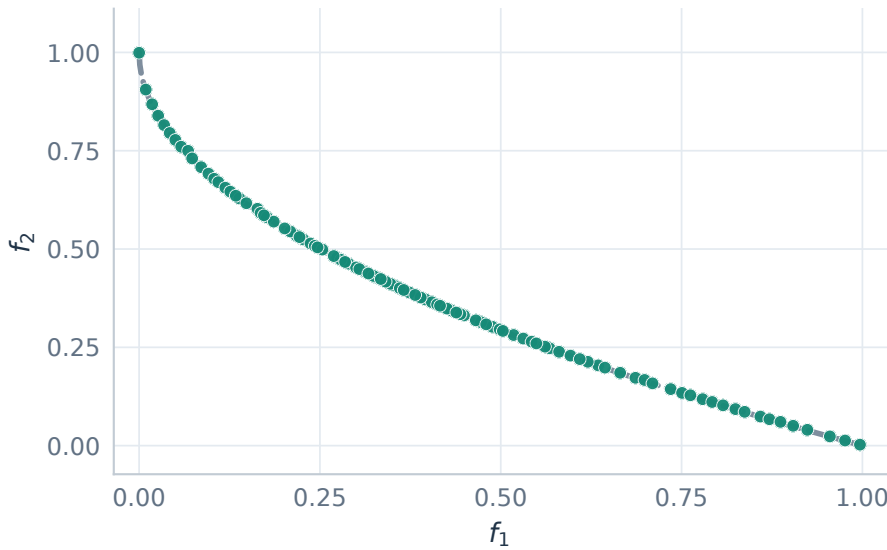
ZDT1 / RVEA*

100 nondominated points | IGD+ 0.003824



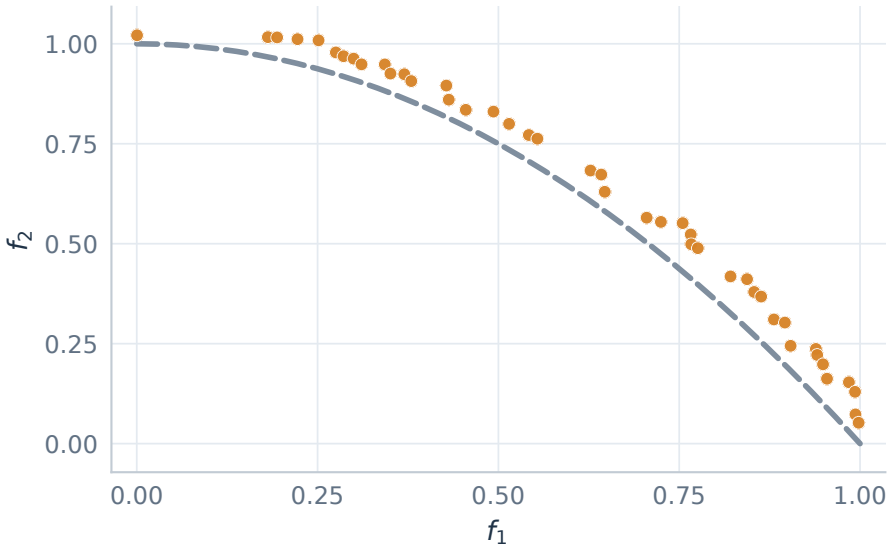
ZDT1 / iRVEA

100 nondominated points | IGD+ 0.003330



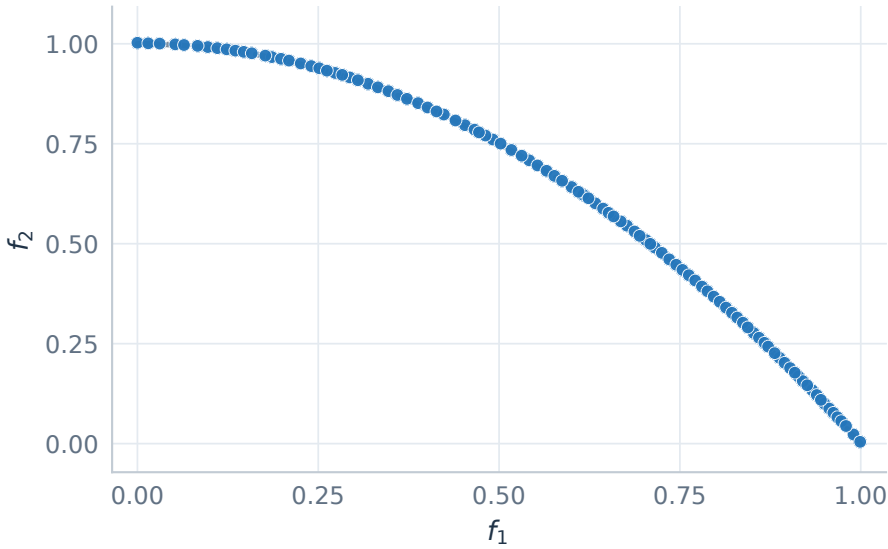
ZDT2 / RVEA

44 nondominated points | IGD+ 0.043022



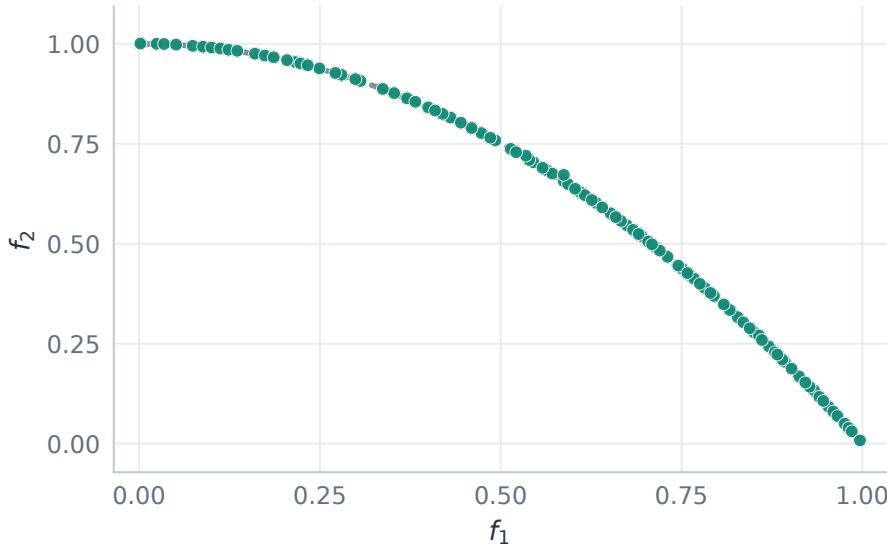
ZDT2 / RVEA*

100 nondominated points | IGD+ 0.003357



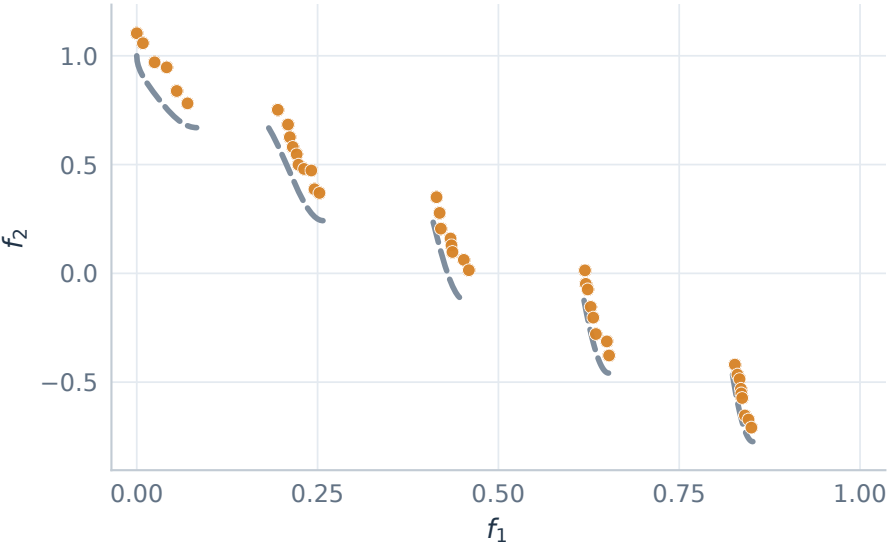
ZDT2 / iRVEA

100 nondominated points | IGD+ 0.003046



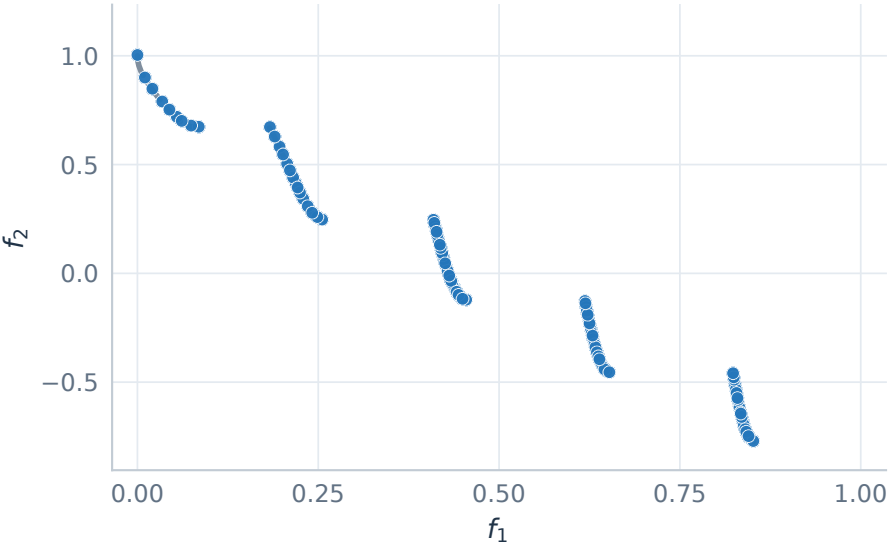
ZDT3 / RVEA

41 nondominated points | IGD+ 0.054089



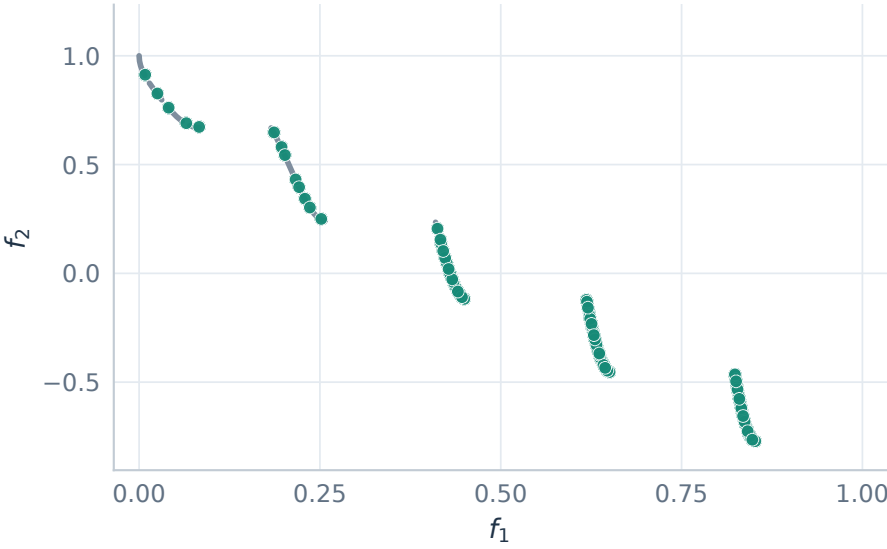
ZDT3 / RVEA*

100 nondominated points | IGD+ 0.003358



ZDT3 / iRVEA

93 nondominated points | IGD+ 0.004571

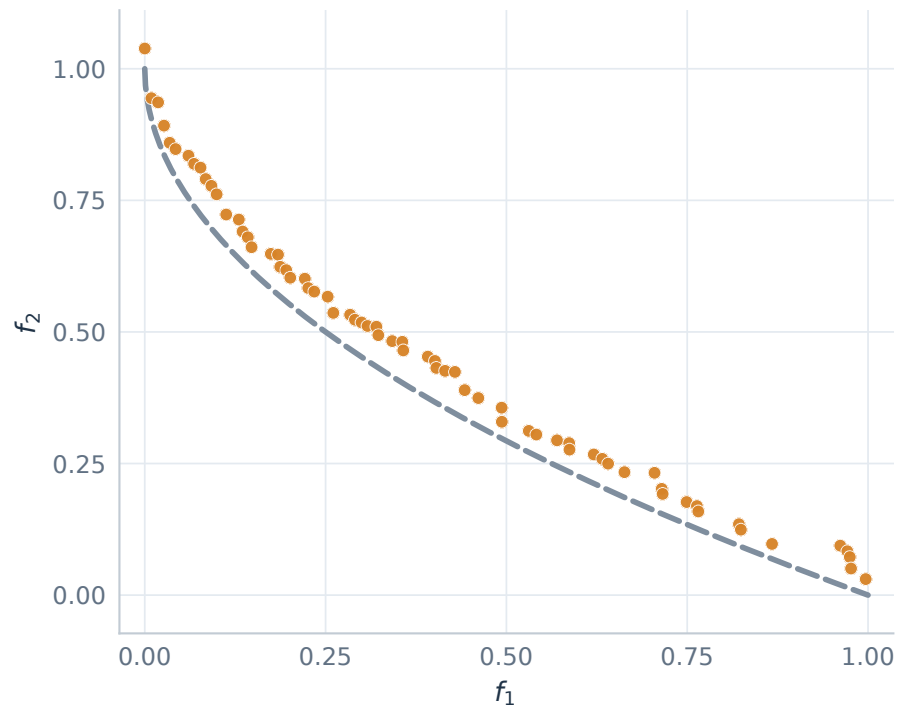


ZDT1 / convex front

Two objectives, 30 variables. Identical axes across algorithms; lower IGD+ is better.

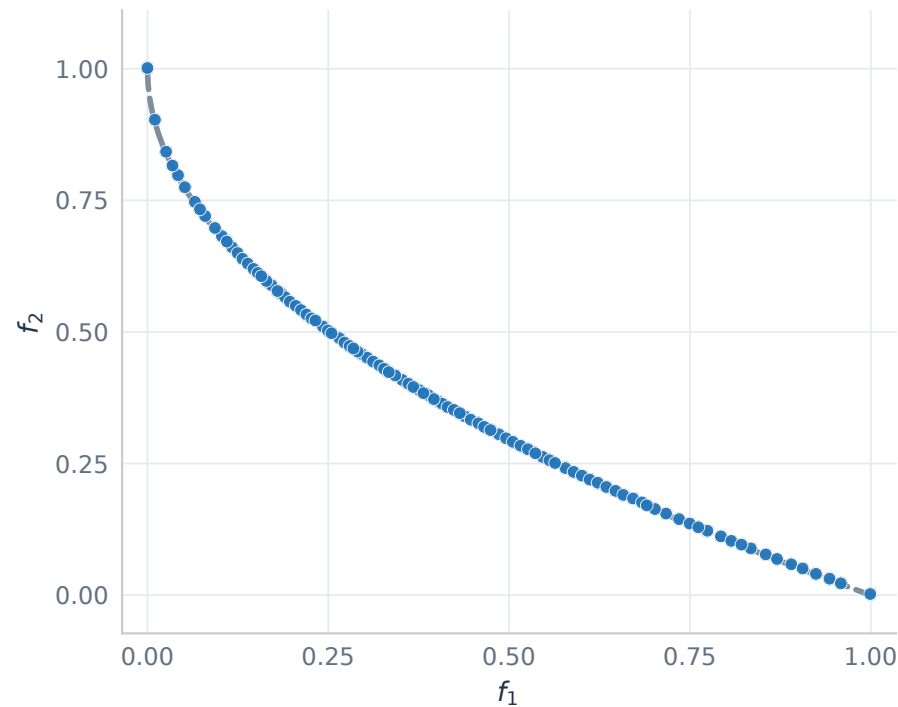
ZDT1 / RVEA

68 nondominated points | IGD+ 0.038176



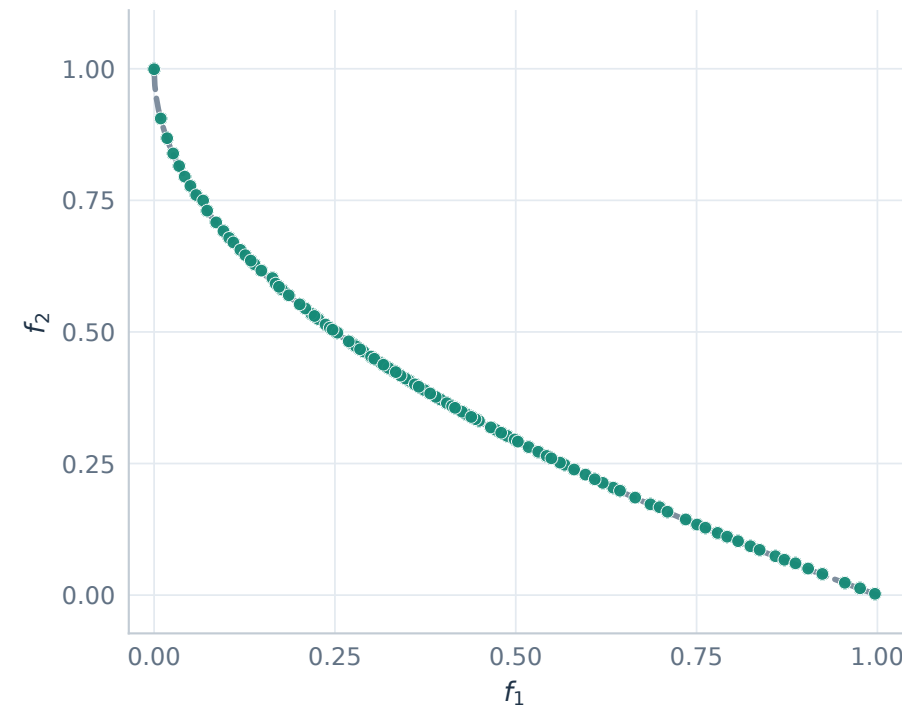
ZDT1 / RVEA*

100 nondominated points | IGD+ 0.003824



ZDT1 / iRVEA

100 nondominated points | IGD+ 0.003330



Seed 42 | 25,000 evaluations | Initial population 100 | Single runs

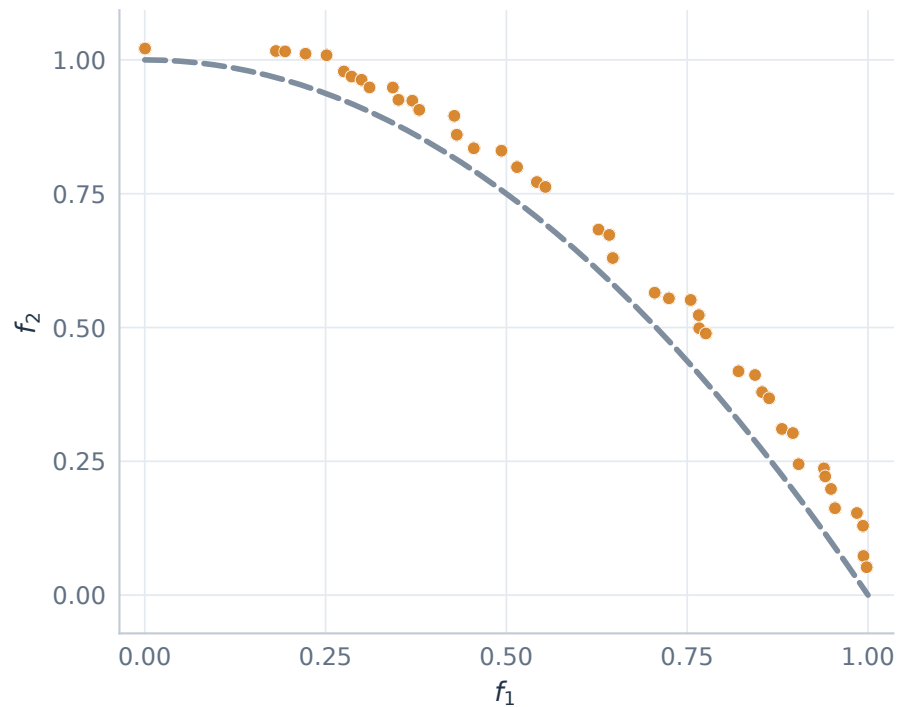
— Reference Pareto front ● Colored points: obtained solutions

ZDT2 / concave front

Two objectives, 30 variables. Identical axes across algorithms; lower IGD+ is better.

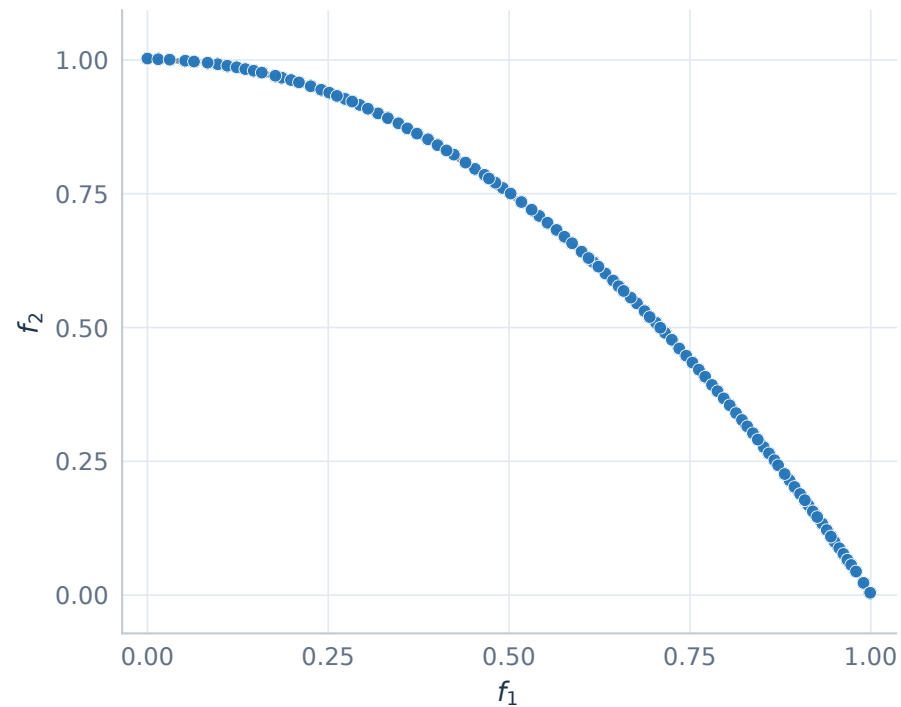
ZDT2 / RVEA

44 nondominated points | IGD+ 0.043022



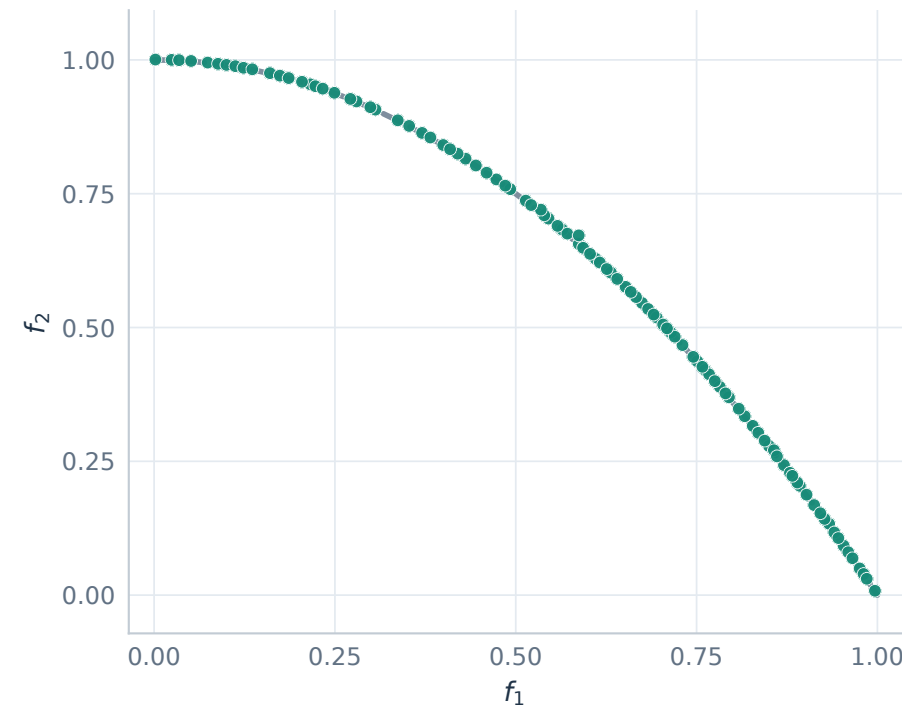
ZDT2 / RVEA*

100 nondominated points | IGD+ 0.003357



ZDT2 / iRVEA

100 nondominated points | IGD+ 0.003046



Seed 42 | 25,000 evaluations | Initial population 100 | Single runs

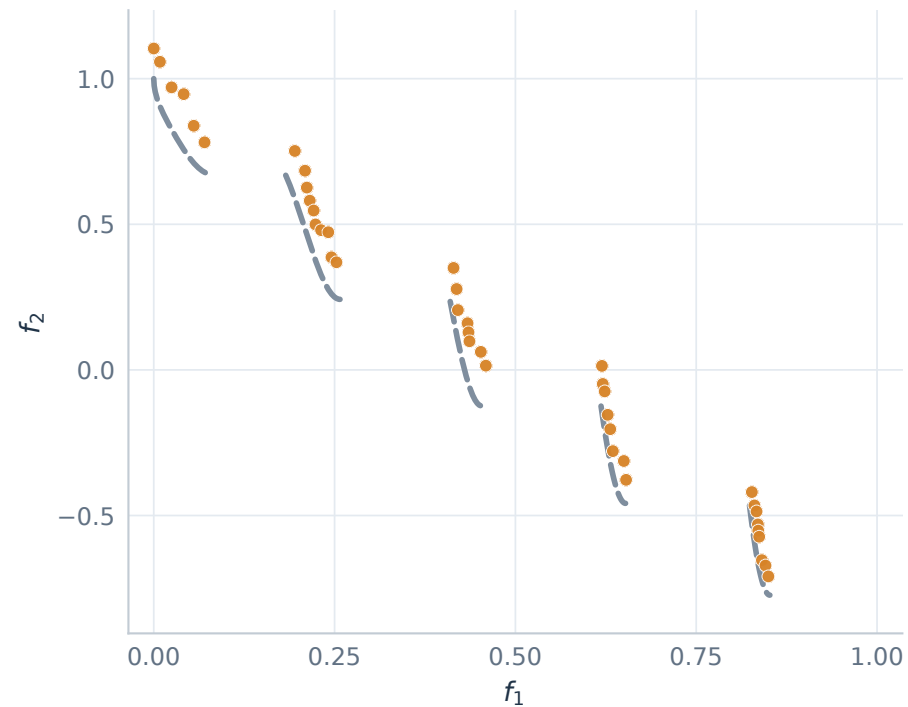
— Reference Pareto front ● Colored points: obtained solutions

ZDT3 / disconnected front

Two objectives, 30 variables. Identical axes across algorithms; lower IGD+ is better.

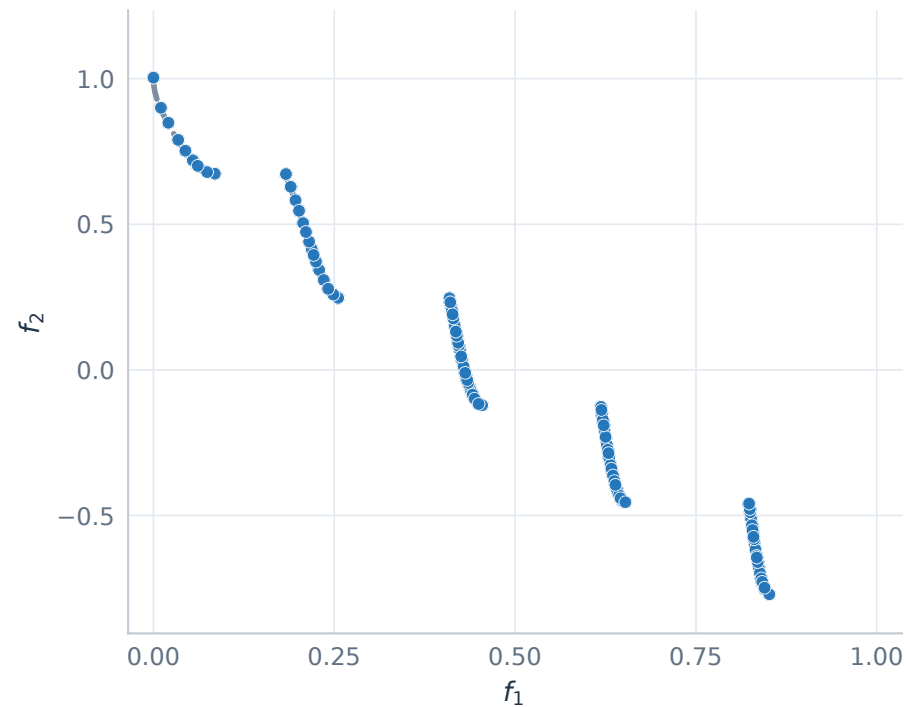
ZDT3 / RVEA

41 nondominated points | IGD+ 0.054089



ZDT3 / RVEA*

100 nondominated points | IGD+ 0.003358



ZDT3 / iRVEA

93 nondominated points | IGD+ 0.004571

